Topic: Laplace transforms for higher order equations

Question: Use a Laplace transform to find the solution to the third order differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$.

$$y''' - 6y'' + 11y' - 6y = e^{4t}$$

Answer choices:

A $$y(t) = -\frac{1}{6}e^{t} + \frac{1}{2}e^{2t} - \frac{1}{2}e^{3t} + \frac{1}{6}e^{4t}$$

B $$y(t) = \frac{1}{8}e^{t} + \frac{1}{4}e^{2t} + \frac{1}{16}e^{3t} + \frac{1}{2}e^{4t}$$

C $$y(t) = 1 + 2e^{t} + 3e^{2t} + 4e^{8t}$$

D $$y(t) = -\frac{1}{3}e^{-3t} + \frac{1}{9}e^{-t} - \frac{1}{9}e^{t} + \frac{1}{3}e^{3t}$$

© Krista King Math

Solution: A

Apply the transform to both sides of the differential equation.

$$\mathscr{L}(y''') - 6\mathscr{L}(y'') + 11\mathscr{L}(y') - 6\mathscr{L}(y) = \mathscr{L}(e^{4t})$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) - 6(s^2Y(s) - sy(0) - y'(0))$$

$$+11(sY(s) - y(0)) - 6Y(s) = \frac{1}{s-4}$$

Substitute the initial conditions $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$,

$$s^3Y(s) - 6s^2Y(s) + 11sY(s) - 6Y(s) = \frac{1}{s-4}$$

then solve for $Y(s)$.

$$(s^3 - 6s^2 + 11s - 6)Y(s) = \frac{1}{s-4}$$

$$Y(s) = \frac{1}{(s-4)(s^3 - 6s^2 + 11s - 6)}$$

$$Y(s) = \frac{1}{(s-1)(s-2)(s-3)(s-4)}$$

Apply a partial fractions decompositions.

$$\frac{1}{(s-1)(s-2)(s-3)(s-4)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3} + \frac{D}{s-4}$$

Solving the decomposition equation gives $A = -1/6$, $B = 1/2$, $C = -1/2$, and $D = 1/6$, so the Laplace transform becomes

© Krista King Math

$$Y(s) = \frac{-\frac{1}{6}}{s-1} + \frac{\frac{1}{2}}{s-2} + \frac{-\frac{1}{2}}{s-3} + \frac{\frac{1}{6}}{s-4}$$

$$Y(s) = -\frac{1}{6}\left(\frac{1}{s-1}\right) + \frac{1}{2}\left(\frac{1}{s-2}\right) - \frac{1}{2}\left(\frac{1}{s-3}\right) + \frac{1}{6}\left(\frac{1}{s-4}\right)$$

Applying the inverse transform to each term gives the general solution to the differential equation.

$$y(t) = -\frac{1}{6}e^{t} + \frac{1}{2}e^{2t} - \frac{1}{2}e^{3t} + \frac{1}{6}e^{4t}$$

© Krista King Math

Topic: Laplace transforms for higher order equations

Question: Use a Laplace transform to find the solution to the third order differential equation, given $y(0) = 2$, $y'(0) = -1$, and $y''(0) = -3$.

$$y''' + y'' + 4y' + 4y = 0$$

Answer choices:

A $y(t) = 2e^{t} + 4e^{6t} + \cos(2t) - \sin(2t)$

B $y(t) = e^{2t} - \sin(2t)$

C $y(t) = e^{-t} + e^{2it}$

D $y(t) = e^{-t} + \cos(2t)$

© Krista King Math

Solution: D

Apply the transform to both sides of the differential equation.

$$\mathscr{L}(y''') + \mathscr{L}(y'') + 4\mathscr{L}(y') + 4\mathscr{L}(y) = \mathscr{L}(0)$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) + s^2Y(s) - sy(0) - y'(0)$$

$$+4(sY(s) - y(0)) + 4Y(s) = 0$$

$$s^3Y(s) - 2s^2 + s + 3 + s^2Y(s) - 2s + 1 + 4(sY(s) - 2) + 4Y(s) = 0$$

$$(s^3 + s^2 + 4s + 4)Y(s) - 2s^2 - s - 4 = 0$$

$$(s^3 + s^2 + 4s + 4)Y(s) = 2s^2 + s + 4$$

Solve for $Y(s)$.

$$Y(s) = \frac{2s^2 + s + 4}{s^3 + s^2 + 4s + 4}$$

$$Y(s) = \frac{2s^2 + s + 4}{(s + 1)(s^2 + 4)}$$

Apply a partial fractions decomposition.

$$\frac{2s^2 + s + 4}{(s + 1)(s^2 + 4)} = \frac{A}{s + 1} + \frac{Bs + C}{s^2 + 4}$$

Solving the decomposition equation gives $A = 1$, $B = 1$, and $C = 0$, so the Laplace transform becomes

$$Y(s) = \frac{1}{s + 1} + \frac{1}{s^2 + 4}$$

© Krista King Math



Applying the inverse transform to each term gives the general solution to the differential equation.

$$y(t) = e^{-t} + \cos(2t)$$

© Krista King Math

Topic: Laplace transforms for higher order equations

Question: Use a Laplace transform to find the solution to the third order differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$.

$$y''' + 2y'' + 4y' + 8y = te^{2t}$$

Answer choices:

A $$y(t) = -\frac{3}{64}e^{2t} + \frac{3}{32}te^{2t} + \frac{23}{64}e^{-2t} + \frac{7}{32}\cos(2t) - \frac{9}{128}\sin(2t)$$

B $$y(t) = \frac{7}{128}e^{t} + \frac{1}{12}te^{3t} + \frac{8}{19}e^{-t} + \frac{7}{32}\cos(4t) + \frac{7}{64}\sin(4t)$$

C $$y(t) = -\frac{3}{128}e^{2t} + \frac{1}{32}te^{2t} + \frac{17}{128}e^{-2t} - \frac{7}{64}\cos(2t) + \frac{9}{64}\sin(2t)$$

D $$y(t) = -\frac{7}{64}e^{2t} + \frac{1}{64}te^{2t} + \frac{34}{128}e^{-2t} - \frac{14}{128}\cos(2t) + \frac{11}{64}\sin(2t)$$

© Krista King Math

Solution: C

Apply the transform to both sides of the differential equation.

$$\mathscr{L}(y''') + 2\mathscr{L}(y'') + 4\mathscr{L}(y') + 8\mathscr{L}(y) = \mathscr{L}(te^{2t})$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) + 2(s^2Y(s) - sy(0) - y'(0))$$

$$+4(sY(s) - y(0)) + 8Y(s) = \frac{1}{(s-2)^2}$$

$$s^3Y(s) - 1 + 2s^2Y(s) + 4sY(s) + 8Y(s) = \frac{1}{(s-2)^2}$$

$$(s^3 + 2s^2 + 4s + 8)Y(s) = \frac{1}{(s-2)^2} + 1$$

$$(s^3 + 2s^2 + 4s + 8)Y(s) = \frac{s^2 - 4s + 5}{(s-2)^2}$$

Solve for $Y(s)$.

$$Y(s) = \frac{s^2 - 4s + 5}{(s-2)^2(s^3 + 2s^2 + 4s + 8)}$$

$$Y(s) = \frac{s^2 - 4s + 5}{(s-2)^2(s+2)(s^2+4)}$$

Apply a partial fractions decomposition.

$$\frac{s^2 - 4s + 5}{(s-2)^2(s+2)(s^2+4)} = \frac{A}{s-2} + \frac{B}{(s-2)^2} + \frac{C}{s+2} + \frac{Ds+E}{s^2+4}$$

© Krista King Math

Solving the decomposition equation gives $A = -3/128$, $B = 1/32$, $C = 17/128$, $D = -7/64$, and $E = 9/32$, so the Laplace transform becomes

$$Y(s) = \frac{-\frac{3}{128}}{s-2} + \frac{\frac{1}{32}}{(s-2)^2} + \frac{\frac{17}{128}}{s+2} + \frac{-\frac{7}{64}s + \frac{9}{32}}{s^2+4}$$

$$Y(s) = \frac{-\frac{3}{128}}{s-2} + \frac{\frac{1}{32}}{(s-2)^2} + \frac{\frac{17}{128}}{s+2} + \frac{-\frac{7}{64}s}{s^2+4} + \frac{\frac{9}{32}}{s^2+4}$$

Applying the inverse transform to each term gives the general solution to the differential equation.

$$y(t) = -\frac{3}{128}e^{2t} + \frac{1}{32}te^{2t} + \frac{17}{128}e^{-2t} - \frac{7}{64}\cos(2t) + \frac{9}{64}\sin(2t)$$

© Krista King Math